METRIC
A-A-59940/1A
September 8, 2021
SUPERSEDING
A-A-59940/1
November 19, 2013

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

CONNECTORS, FIBER OPTIC, SINGLE FIBER, SMALL FORM FACTOR, LC TYPE

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

The complete requirements for procuring LC connectors described herein shall consist of this document and the latest issue in effect of A-A-59940.

CLASSIFICATION: LC connectors are of the termination processes in table I, the connector types in table II, the temperature ranges in table III, the housing type configurations in table IV, and the strain relief designs in table V.

TABLE I. Termination process.

Termination process	Designation
Epoxy polish	1
Quick-connect – fusion splice	2
Quick-connect – mechanical splice	3

TABLE II. Connector type.

Fiber cable class	Designation
MM – UPC (graded-index, glass core, and glass cladding, multimode)	1 (or MM)
SM – UPC (dispersion-unshifted, glass core, and glass cladding, single-mode)	2 (or SM)
SM – APC (dispersion-unshifted, glass core, and glass cladding, single-mode)	3 (or APC)

Beneficial comments, recommendations, additions, deletions, clarifications, and any other data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 6060

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TABLE III. Temperature ranges.

Temperature range designation	Operating temperature	Non-operating temperature	Storage temperature
	°C (°F)	°C (°F)	°C (°F)
1	-28 to +65 (-18.4 to +149)	-40 to +70 (-40 to +158)	-40 to +70 (-40 to +158)
2	-28 to +85 (-18.4 to +185)	-40 to +70 (-40 to +158)	-40 to +70 (-40 to +158)
3	-55 to +125 (-67 to +257)	-55 to +125 (-67 to +257)	-55 to +125 (-67 to +257)

TABLE IV. Housing type configurations.

Fiber cable class	Designation
Simplex – one optical fiber/ferrule/single ferrule	S
Duplex – one optical fiber/ferrule/two ferrules	D

TABLE V. Strain relief.

Strain relief design	Designation
Pull-proof 1/	P
Non-pull-proof 2/	N

NOTES:

1/ A connector with pull-proof type strain relief is designed to be used with cable considered to be of non-tight construction. This includes furcation units in accordance with A-A-59729 (2 to 2.4 millimeters in diameter).

2/ A connector with non-pull-proof type strain relief is designed to be used with cable considered to be of tight construction. This includes cable in accordance with MIL-PRF-85045/16 (2 millimeters in diameter).

Part or Identification Number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

AA59940	**/01**	**-**	**X**	**X**	**X**	**X**	**X**
CID number	Specification sheet		Termination process (1, 2, or 3) (see table I)	Connector type (1, 2, or 3) (see table II)	Temperature range (1, 2, or 3) (see table III)	Housing (S or D) (see table IV)	Strain relief (N or P) (see table V)

PIN example: AA59940/01-111SN is an LC connector for buffered fiber and tight structured OFCC, multimode fiber, simplex housing, operating temperature range of -28 to +65 °C (-18.4 to +149 °F), ultra physical contact polish, and epoxy polish termination method.

For connector configurations, see tables VI through VIII.

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TABLE VI. Connector code configuration number: termination process: epoxy polish.

Connector configuration number	Termination process (see table I)	Connector type (see table II)	Temperature range (see table III)	Housing (see table IV)	Strain relief (see table V)
111 SN	1	MM – UPC (1)	1	Simplex (S)	N
111 SP					P
121 SN		SM – UPC (2)			N
121 SP					P
111 DN		MM – UPC (1)		Duplex (D)	N
111 DP					P
121 DN		SM – UPC (2)			N
121 DP					P
112 SN	1	MM – UPC (1)	2	Simplex (S)	N
112 SP					P
122 SN		SM – UPC (2)			N
122 SP					P
112 DN		MM – UPC (1)		Duplex (D)	N
112 DP					P
122 DN		SM – UPC (2)			N
122 DP					P
113 SN	1	MM – UPC (1)	3	Simplex (S)	N
113 SP					P
123 SN		SM – UPC (2)			N
123 SP					P
113 DN		MM – UPC (1)		Duplex (D)	N
113 DP					P
123 DN		SM – UPC (2)			N
123 DP					P

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TABLE VII. Connector code configuration number: termination process: fusion splice.

Connector configuration number	Termination process (see table I)	Connector type (see table II)	Temperature range (see table III)	Housing (see table IV)	Strain relief (see table V)
211SN	2	MM – UPC (1)	1	Simplex (S)	N
211 SP					P
221 SN		SM – UPC (2)			N
221 SP					P
231 SN		SM – APC (3)			N
231 SP					P
211 DN		MM – UPC (1)		Duplex (D)	N
211 DP					P
221 DN		SM – UPC (2)			N
221 DP					P
231 DN		SM – APC (3)			N
231 DP					P
212 SN	2	MM – UPC (1)	2	Simplex (S)	N
212 SP					P
222 SN		SM – UPC (2)			N
222 SP					P
232 SN		SM – APC (3)			N
232 SP					P
212 DN		MM – UPC (1)		Duplex (D)	N
212 DP					P
222 DN		SM – UPC (2)			N
222 DP					P
232 DN		SM – APC (3)			N
232 DP					P
213 SN	2	MM – UPC (1)	3	Simplex (S)	N
213 SP					P
223 SN		SM – UPC (2)			N
223 SP					P
233 SN		SM – APC (3)			N
233 SP					P
213 DN		MM – UPC (1)		Duplex (D)	N
213 DP					P
223 DN		SM – UPC (2)			N
223 DP					P
233 DN		SM – APC (3)			N
233 DP					P

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TABLE VIII. Connector code configuration number: termination process: mechanical splice.

Connector configuration number	Termination process (see table I)	Connector type (see table II)	Temperature range (see table III)	Housing (see table IV)	Strain relief (see table V)
311 SN	3	MM – UPC (1)	1	Simplex (S)	N
311 SP					P
321 SN		SM – UPC (2)			N
321 SP					P
331 SN		SM – APC (3)			N
331 SP					P
311 DN		MM – UPC (1)		Duplex (D)	N
311 DP					P
321 DN		SM – UPC (2)			N
321 DP					P
331 DN		SM – APC (3)			N
331 DP					P
312 SN	3	MM – UPC (1)	2	Simplex (S)	N
312 SP					P
322 SN		SM – UPC (2)			N
322 SP					P
332 SN		SM – APC (3)			N
332 SP					P
312 DN		MM – UPC (1)		Duplex (D)	N
312 DP					P
322 DN		SM – UPC (2)			N
322 DP					P
332 DN		SM – APC (3)			N
332 DP					P
313 SN	3	MM – UPC (1)	3	Simplex (S)	N
313 SP					P
323 SN		SM – UPC (2)			N
323 SP					P
333 SN		SM – APC (3)			N
333 SP					P
313 DN		MM – UPC (1)		Duplex (D)	N
313 DP					P
323 DN		SM – UPC (2)			N
323 DP					P
333 DN		SM – APC (3)			N
333 DP					P

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DESIGN AND CONSTRUCTION

Interface dimensions. The interface dimensions of the LC connector shall be in accordance with TIA-604-10 (Fiber Optic Connector Intermateability Standard [FOCIS 10]). For additional dimensions, see figure 1, table IX, and figure 2.

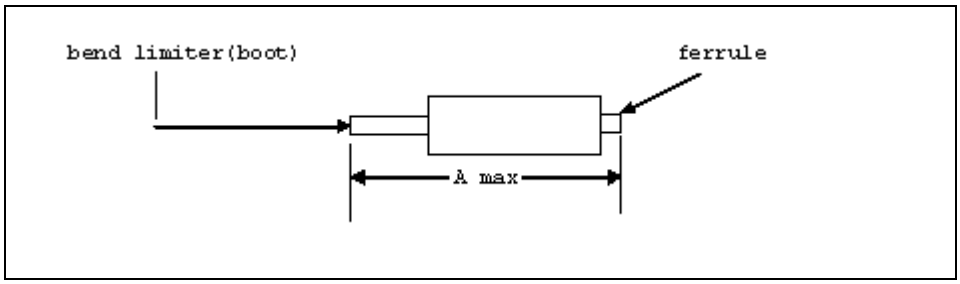


FIGURE 1. Size.

TABLE IX. Fiber/cable type compatibility. 1/

Dimension	Buffered fiber (250/900 µm) 2/	OFCC – tight structure (2 mm) 2/	OFCC – non-tight structure (2 mm) 2/
A (max)	32 mm	56.3 mm	55 mm

NOTES:
1/ See table VI.
2/ See A-A-59940 for additional information.

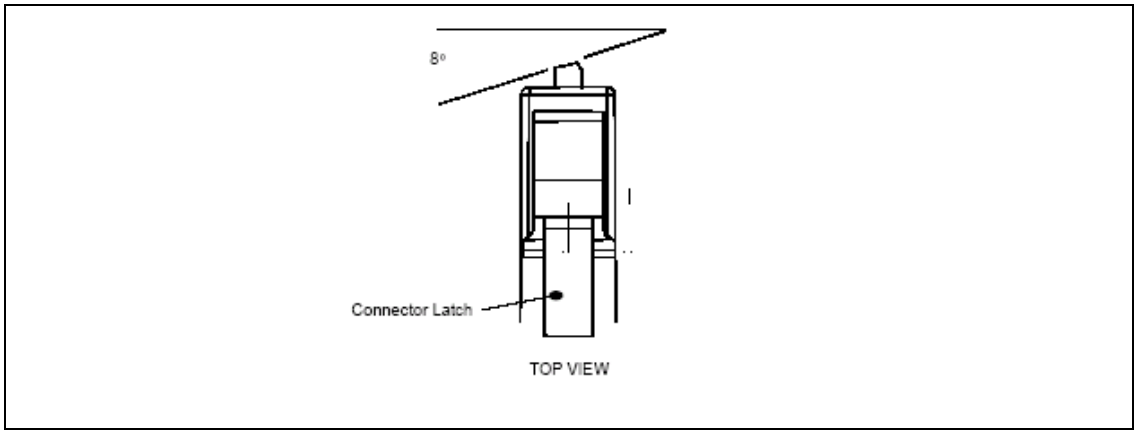


FIGURE 2. LC APC ferrule/angle orientation.

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Insertion loss. For insertion loss requirements, see table X.

TABLE X. Performance requirements.

Requirement	MM – UPC	SM – UPC	SM – APC
Initial insertion loss (dB)	0.50	0.50	0.50
Insertion loss verification (dB)	0.75	0.75	0.75
Return loss (dB)	1/	30/40/45 2/	≥58
Color (part of housing or boot)	Beige	Blue or yellow	Green
Endface geometry	3/	3/	Telcordia GR-326-CORE, or equal

NOTES:
1/ Not applicable.
2/ 30 dB applies to epoxy polish and standard polish connectors, 40 dB applies to epoxy polish enhanced polish connectors, and 45 dB applies to quick-connect connectors.
3/ Applicable.

Return loss. For return loss requirements, see table X.

Color. For color requirements, see table X.

Endface geometry. For endface geometry requirements, see table X.

Cable pull-out (fiber cable connector configuration B, C, and D only). The connector shall be capable of sustaining a static tensile load of at least 50 newtons (11.24 pounds) applied for 1 minute to the cable behind the back end of the connector without evidence of physical damage to the connector or the optical fiber cable. The connector shall be mated to a fixed connector adapter for this test. The change in optical transmittance shall not exceed 0.5 decibel after the test. The return loss of the connector shall be as specified after the test.

Crimp tool compatibility. The LC 2-millimeter cordage connector configuration meets all requirements when the crimp sleeve is assembled to the jumper connector using a circular indent ring crimp die with a 2.67±0.08-millimeter (0.105±0.003-inch) minor diameter and a 3.43±0.08-millimeter (0.135±0.003-inch) major diameter or using a hex crimp die with flats measuring 2.947+0.025 millimeters (0.116+0.001 inch) across and 1.25 millimeter (0.049 inch) maximum length.

Fusion splicer compatibility. The LC connector configuration capable of being fusion spliced (see table VII) shall be capable of assembly using fusion splicers approved to A-A-59799, configuration 2.

Mechanical splice tool compatibility. The LC connector configuration capable of being mechanically spliced (see table VII) shall permit fiber preparation and assembly of the splice utilizing tools and supplies available in the standard Navy shipboard fiber optic toolkit, as described in NAVSEA Drawings 499-6872811, 499-6872813, or 499-7085185, supplemented with parts and tools provided with the mechanical splice kit.

Conformance inspection. Test performed, test procedure, connector performance, and test sample (specimen) quantity shall be in accordance with table XI, with any exception (such as a modification or addition) listed in this specification sheet. A duplex version of a particular connector PIN shall be recognized as conforming so long as the simplex version has shown to be conforming and the only change from the conforming simplex connector to the duplex connector is that the housing is duplex.

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TABLE XI. Conformance inspection.

Group	Inspection	A-A-59940 requirement and test paragraph 1/	Specimen 2/		
			Connector	Connector on cable type A	Connector on cable type B or C
I	**Visual and mechanical inspection**				
	Interchangeability and intermateability	3.1.1	1-4	1-8	1-16
	Interface dimensions	3.1.2			
	Cable strain relief, bend limiter	3.1.3, 3.1.4			
	Optical fiber compatibility	3.1.5			
	Optical fiber attachment (ferrule, quick-connect)	3.1.6			
	Color	3.1.7			
	Endface geometry (quick-connect)	3.1.8			
	Endface condition (quick-connect)	3.1.9			
	Cleaved fiber stub endface condition (quick-connect)	3.1.10			
	Ferrule extension and contact force	3.1.11			
	Maintainability (cleaning, field termination, dust cover)	3.6.1, 3.6.2, 3.6.3			
	Size, mass, workmanship, marking	3.7, 3.8, 3.9, 3.10			
II	**Performance**				
	Insertion loss, initial	3.5.1	N/A	1-8	1-16
	Ambient light susceptibility	3.5.2			
	Return loss	3.5.3			
III	**Mechanical properties**				
	Cable pull-out	3.3.1	N/A	N/A	1-8
	Fiber pull-out	3.3.2			N/A
	Fiber-to-ferrule pull-out	3.3.2.1		1-4	N/A
	Fiber-to-splice pull-out	3.3.2.1		6-8	N/A
	Flex	3.3.3		N/A	1-8
	Twist	3.3.4			
	Force to engage/disengage	3.3.5			
	Impact	3.3.6			
	90-degree cable pull-out	3.3.7			
	Strength of coupling mechanism	3.3.8			
	Mating durability	3.3.9			
	Insertion loss verification	3.5.4			
	Shock-COTS	3.3.10			
	Vibration	3.3.11			

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TABLE XI. Conformance inspection – Continued.

Group	Inspection	A-A-59940 requirement and test paragraph 1/	Specimen 2/		
			Connector	Connector on cable type A	Connector on cable type B or C
IV	**Environmental properties**				
	Temperature	3.4.1	N/A	N/A	9-16
	Temperature humidity cycling	3.4.2			
	Temperature cycling	3.4.3			
	Life aging (temperature life)	3.4.4			
	Thermal shock	3.4.5			
	Insertion loss verification	3.5.4			
V	**Materials**				
	Non-metallic materials (adhesives, index matching gel, sealing compounds)	3.2.1	3/	N/A	N/A
	Metallic materials (dissimilar metals)	3.2.2			
	Toxic and hazardous products and formulations	3.2.3			
	Fungus	3.2.4			
	Oxygen index	3.2.5			
	Salt spray	3.2.6			
	Flammability	3.2.7			
	Radioactive materials	3.2.8			
	Lubricants	3.2.9			
	Sealing compounds	3.2.10			
	Liquid materials	3.2.11			
	Ferrule	3.2.12			

NOTES:

1/ Individual item requirements shall be as specified in A-A-59940 and this specification sheet. In the event of any conflict between the requirements of A-A-59940 and this specification sheet, the latter shall govern.

2/ Specimen shall be simplex connectors.

3/ Applicable connector material samples may be provided in lieu of connector samples as test requirement and method require. Material certification may be provided as proof of conformance.

Intended use. LC connectors in accordance with this specification sheet are intended to be used in a sheltered environment with commercial fiber optic cable, fiber optic cable in accordance with MIL-PRF-85045, and blown optical fiber tube furcation units in accordance with A-A-59729.

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CONCLUDING MATERIAL

MILITARY INTERESTS

Custodians:
Army – CR
Navy – SH
Air Force – 85

Review Activities:
Air Force – 13, 19, 93
Misc – DI

Preparing Activity:
Navy – SH
(Project 6060-2021-008)

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